

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper: CS-108 — Data Structures Algorithms

Time: Three hours

Maximum Marks: 70

*Instructions: Attempt any five questions. All questions carry equal marks.***Question 1.**

- (a) Explain primitive and non-primitive data structures with suitable examples.
- (b) Distinguish between arrays and linked lists with suitable illustrations.

Question 2.

- (a) What is a Binary Search Tree (BST)? How is a BST different from a B-Tree? Explain with a suitable example.
- (b) Discuss tree traversal techniques with suitable examples.

Question 3.

- (a) Define Quick Sort. Sort the following array using the Quick Sort algorithm:

$$12, 98, 1, 55, 3, 77, 0, 10$$

- (b) Discuss the time complexity of Quick Sort in the best, average, and worst cases.

Question 4.

- (a) Explain Greedy and Dynamic programming approaches.
- (b) Choose a problem of your choice and solve it using both approaches.

Question 5.

- (a) Define asymptotic notations ($\mathcal{O}$, Ω , Θ) with suitable examples.
- (b) Find the upper, lower, and tight bounds of the running time of the quadratic function:

$$f(n) = 99n^2 + 28n + 4$$

Question 6.

- (a) Define backtracking. Discuss it through a suitable example.
- (b) Define Graph as a data structure. Explain the adjacency matrix and adjacency list representations with suitable examples.

Question 7.

- (a) Define the shortest path problem with suitable illustrations.
- (b) Define Hashing and Collision. What are the methods used to handle collisions? Explain with an example.